

CS 342: Computer Vision and Pattern Recognition

Instructor: Tamal Mj

Shanti Mantra

Lyrics in Sanskrit

Meaning in English

ॐ सह नाववतु।

Om, May we all be protected

सह नौ भुनक्तु।

May we all be nourished

सह वीर्यं करवावहै।

May we work together with great energy

तेजस्यि नावधीतमस्तु

May our intellect be sharpened (may our study be effective)

मा विद्विषावहै।

Let there be no Animosity amongst us

ॐ शान्तिः शान्तिः शान्तिः ॥

Om, peace (in me), peace (in nature), peace (in divine forces)

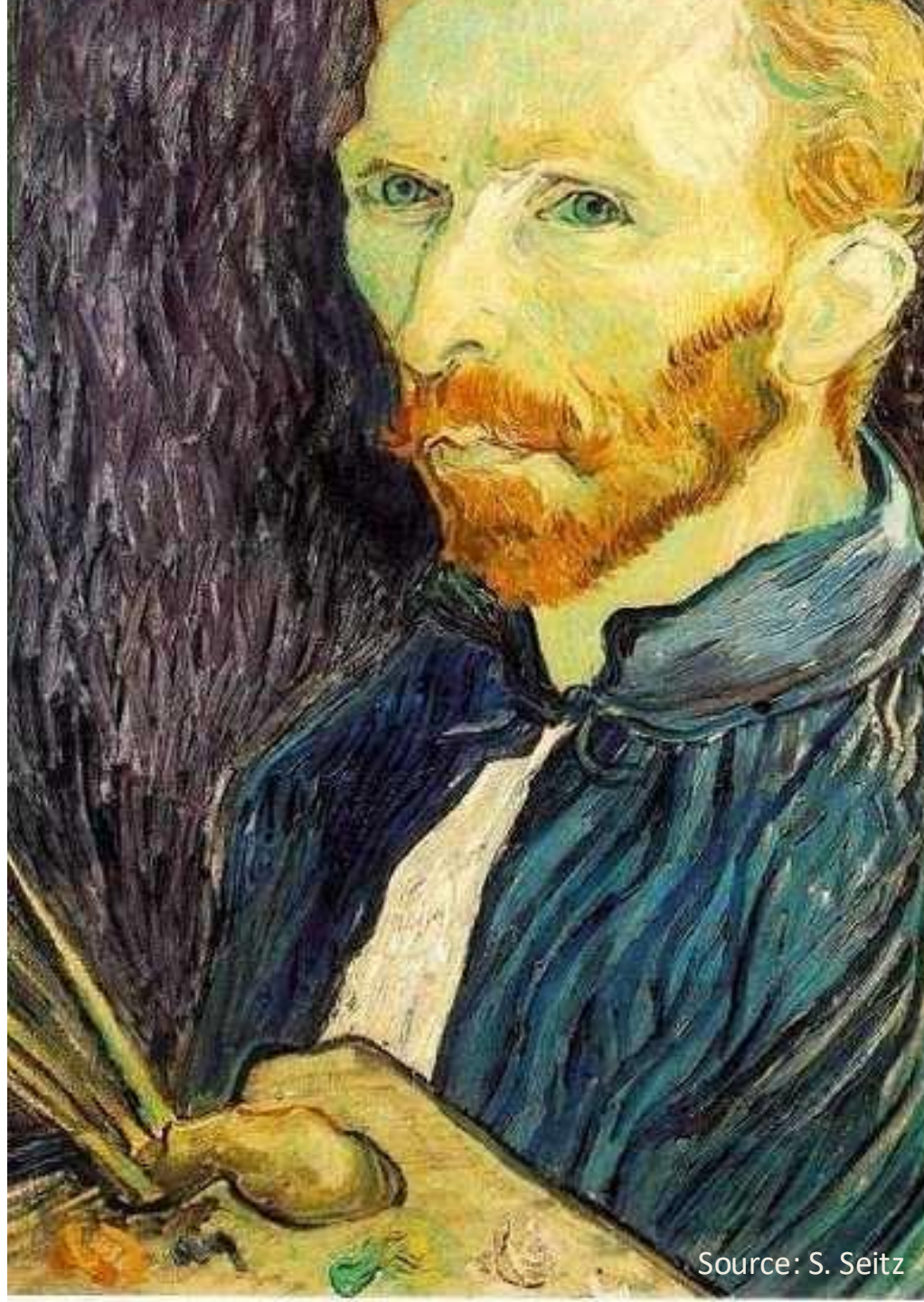
Slides by Noah Snaveley

Image sub-sampling



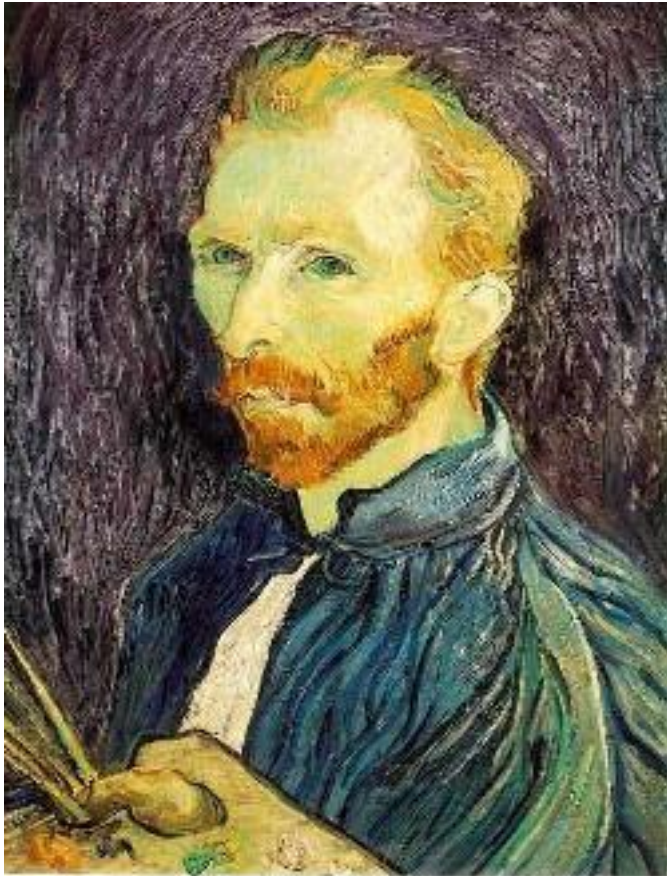
Image Scaling

This image is too big to fit on the screen. How can we generate a half-sized version?



Source: S. Seitz

Image sub-sampling



1/4



1/8

Throw away every other row and column to create a 1/2 size image
- called *image sub-sampling*

Image sub-sampling



$1/2$



$1/4$ (2x zoom)

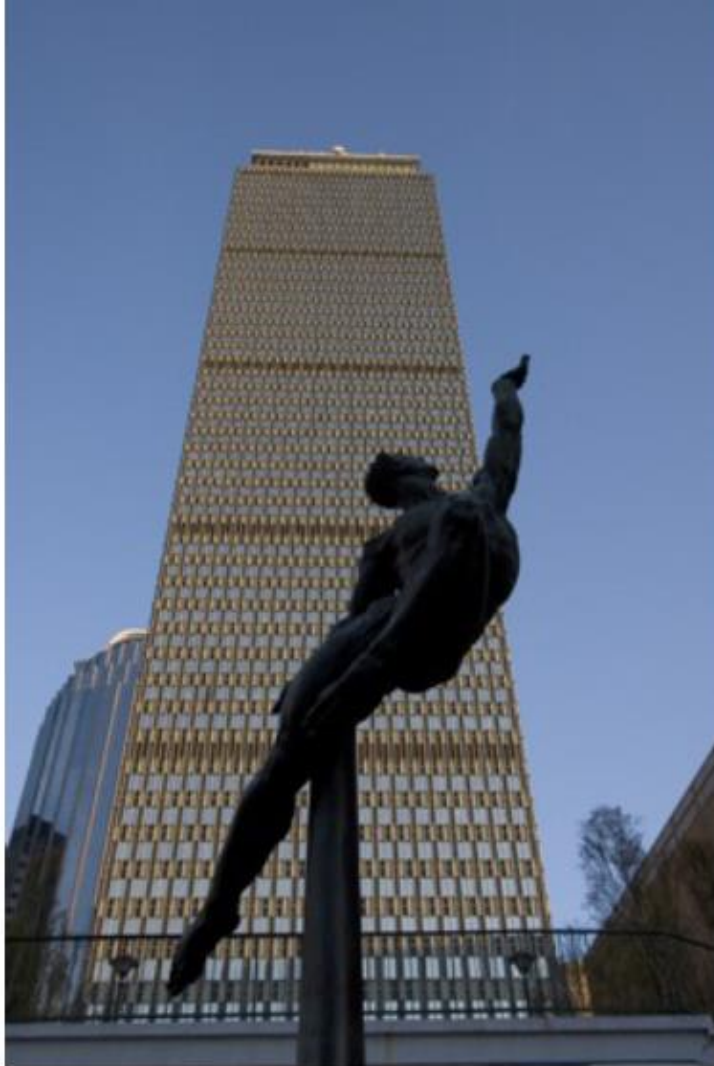


$1/8$ (4x zoom)

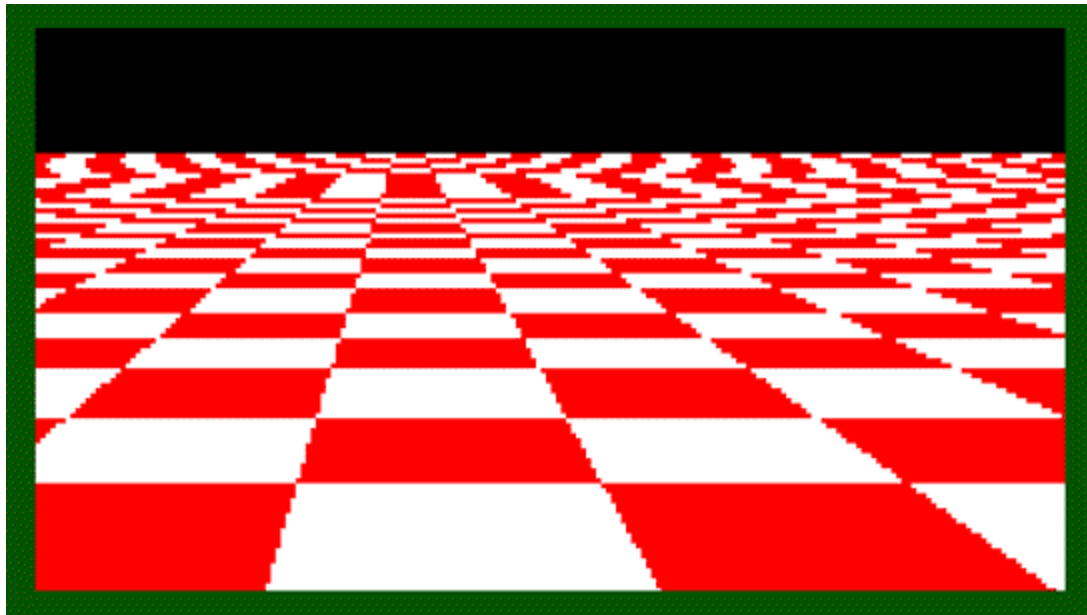
Why does this look so cruffy?

Source: S. Seitz

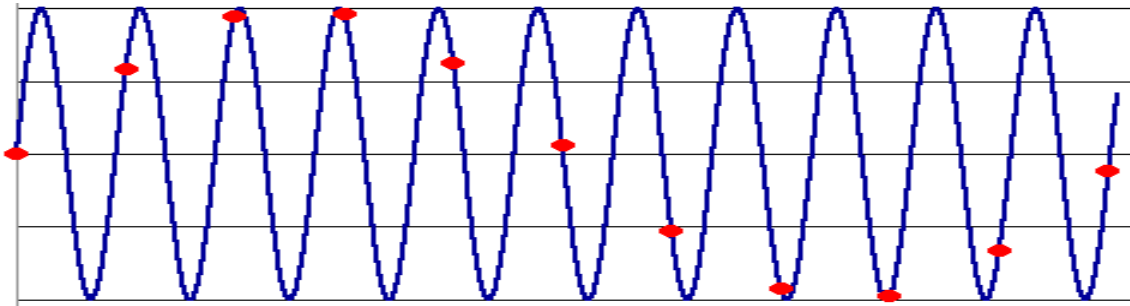
Image sub-sampling



Even worse for synthetic images



Aliasing



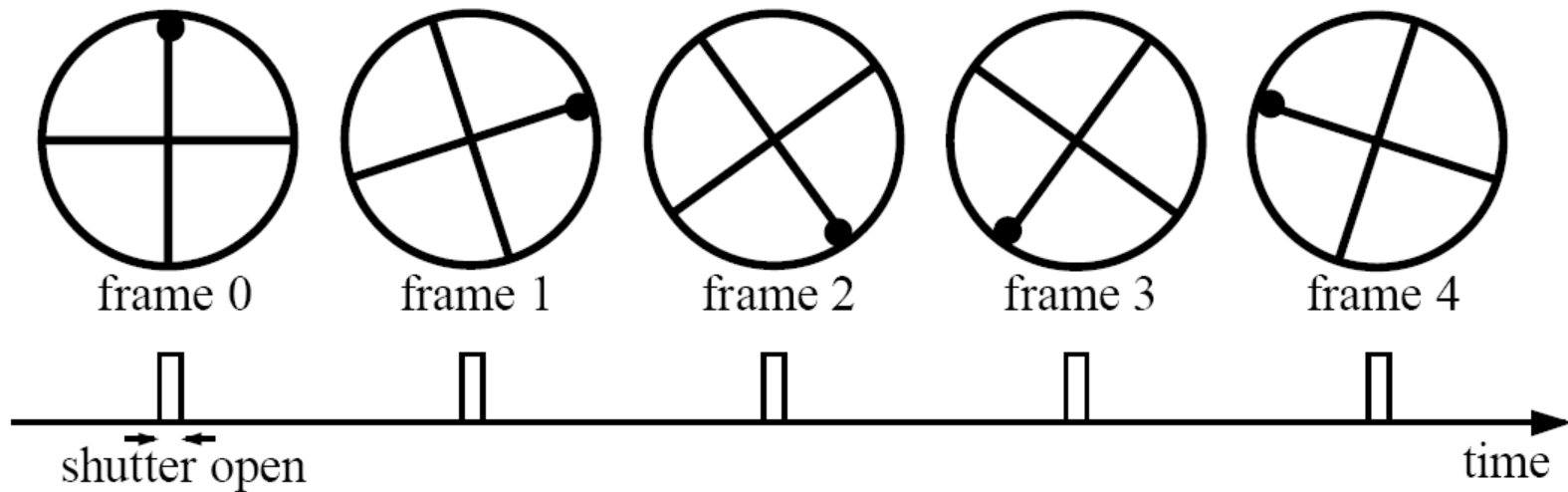
- Occurs when your sampling rate is not high enough to capture the amount of detail in your image
- Can give you the wrong signal/image—an *alias*
- To do sampling right, need to understand the structure of your signal/image
- Enter Monsieur Fourier...
- To avoid aliasing:
 - sampling rate $\geq 2 * \text{max frequency in the image}$
 - said another way: $\geq$ two samples per cycle
 - This minimum sampling rate is called the **Nyquist rate**

Wagon-wheel effect

Imagine a spoked wheel moving to the right (rotating clockwise).

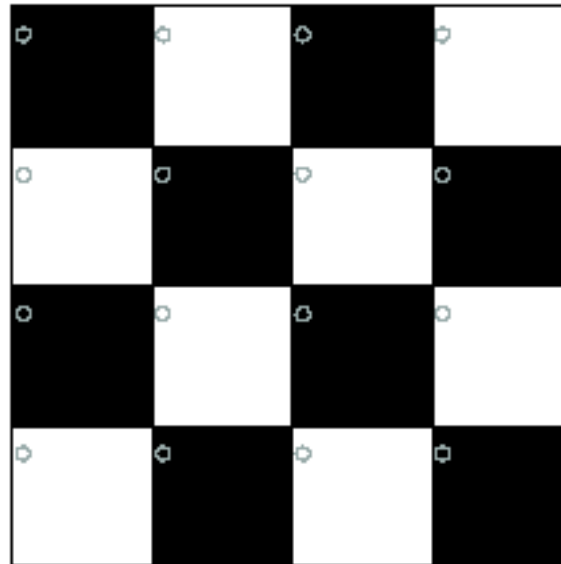
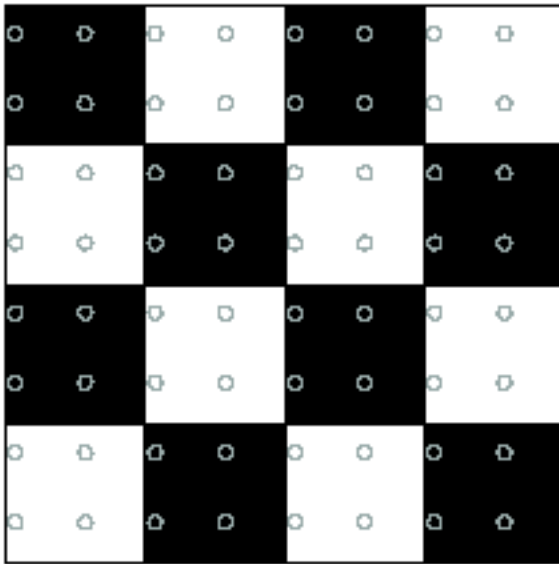
Mark wheel with dot so we can see what's happening.

If camera shutter is only open for a fraction of a frame time (frame time = $1/30$ sec. for video, $1/24$ sec. for film):

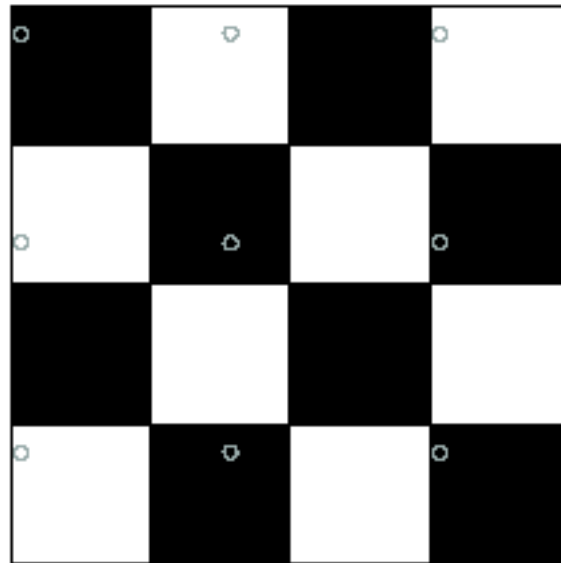
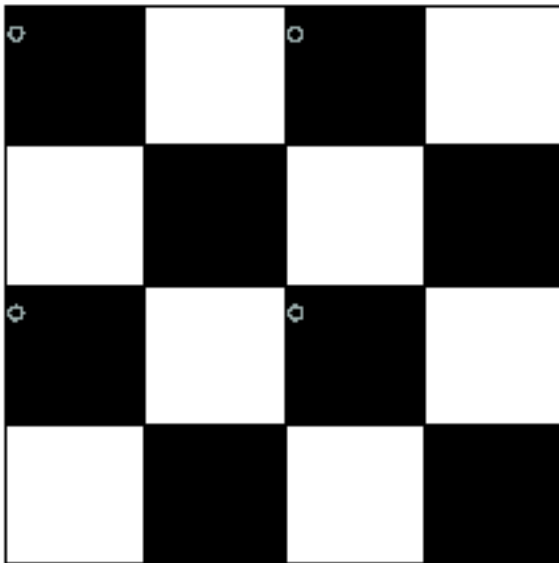


Without dot, wheel appears to be rotating slowly backwards!
(counterclockwise)

Nyquist limit – 2D example



Good sampling

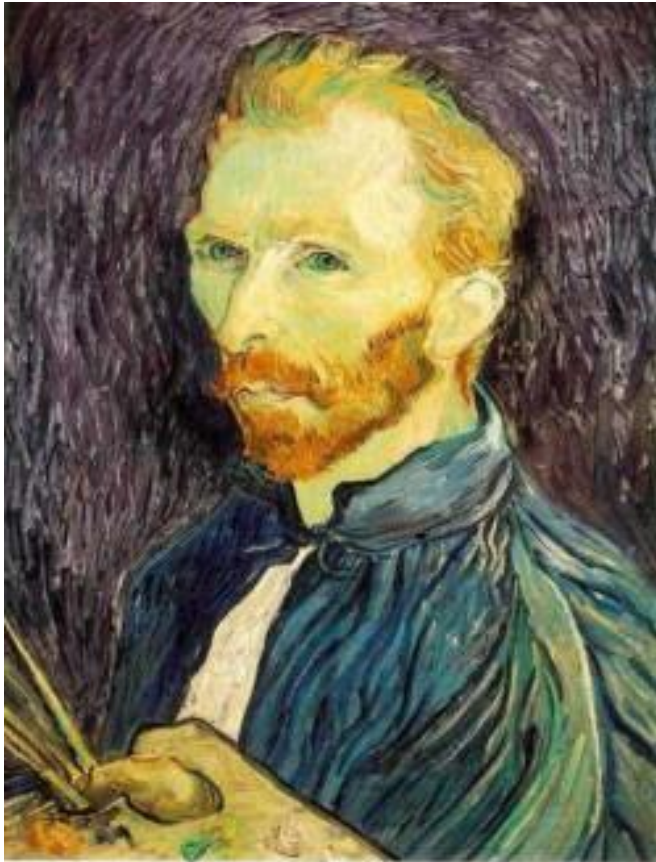


Bad sampling

Aliasing

- When downsampling by a factor of two
 - Original image has frequencies that are too high
- How can we fix this?

Gaussian pre-filtering



Gaussian 1/2



G 1/4



G 1/8

- Solution: filter the image, *then* subsample

Subsampling with Gaussian pre-filtering



Gaussian 1/2



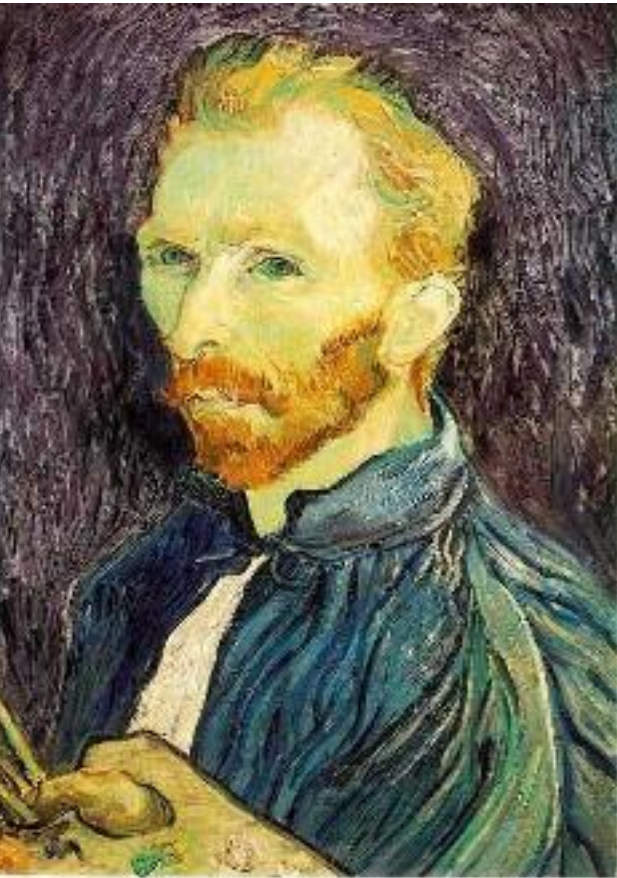
G 1/4



G 1/8

- Solution: filter the image, *then* subsample

Compare with...



1/2



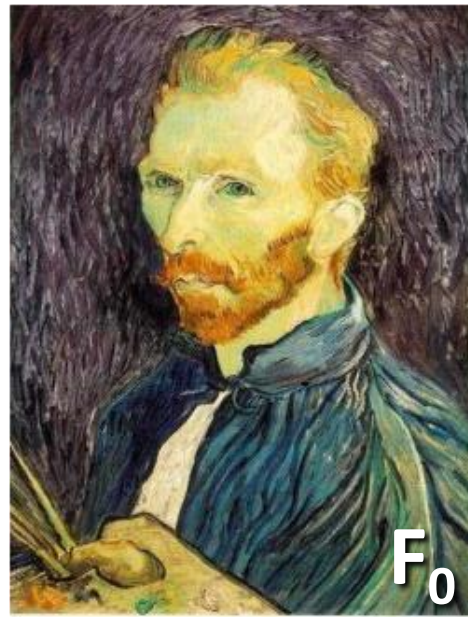
1/4 (2x zoom)



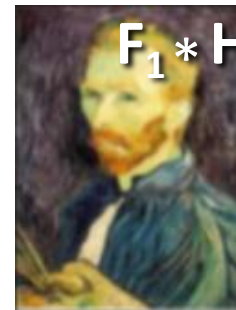
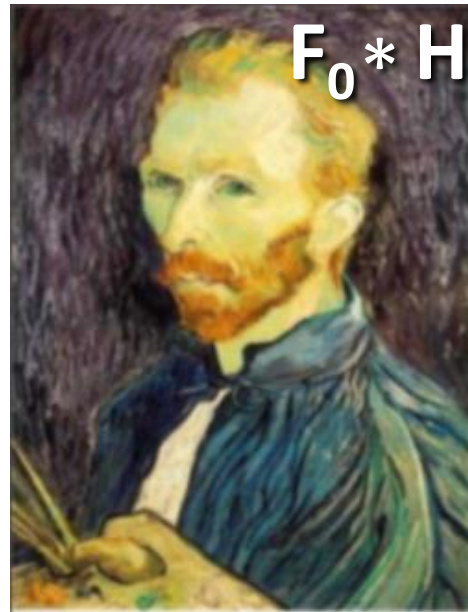
1/8 (4x zoom)

Gaussian pre-filtering

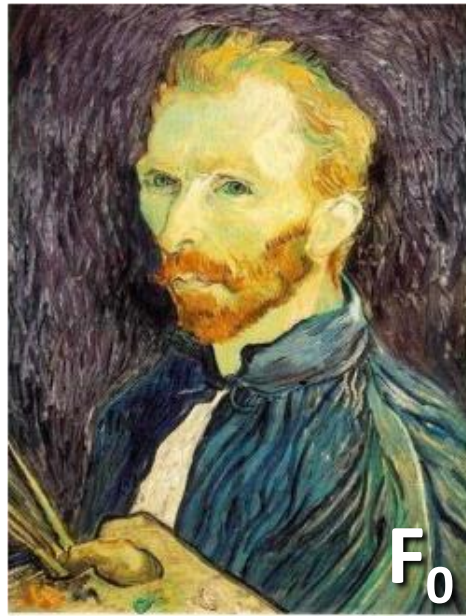
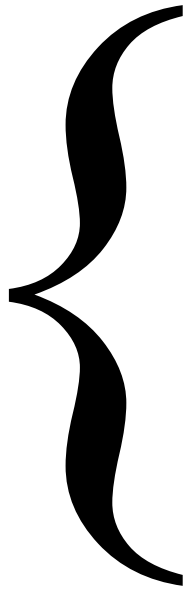
- Solution: filter the image, *then* subsample



...



*Gaussian
pyramid*



F_0



F_1



F_2



blur



subsample

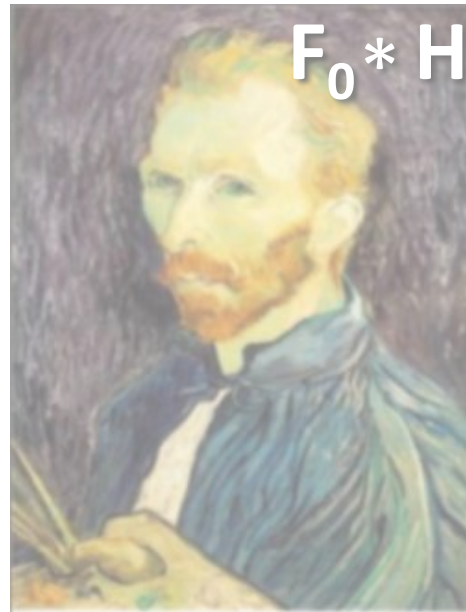


blur

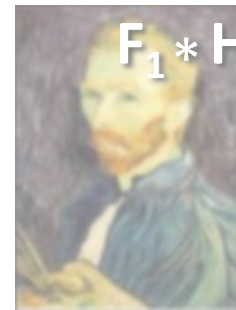


subsample

...



$F_0 * H$



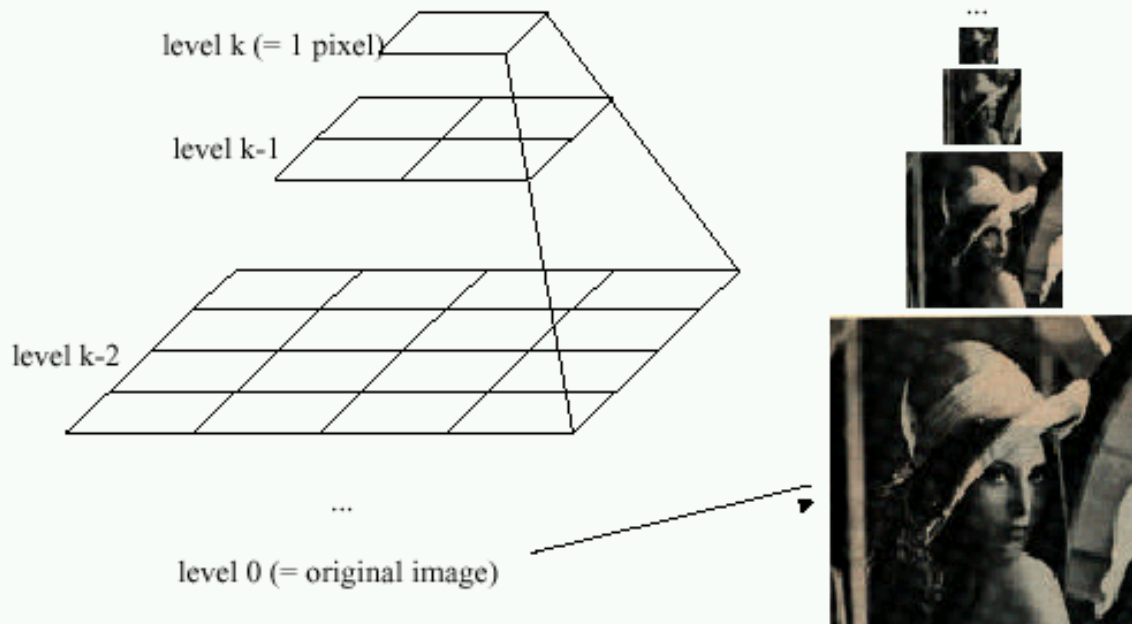
$F_1 * H$



Gaussian pyramids

[Burt and Adelson, 1983]

Idea: Represent $N \times N$ image as a “pyramid” of $1 \times 1, 2 \times 2, 4 \times 4, \dots, 2^k \times 2^k$ images (assuming $N = 2^k$)



Gaussian Pyramids have all sorts of applications in computer vision

Gaussian Pyramid

